SUMMARY

Derivatives	Antiderivatives
$\frac{d}{dx}(c)=0,$ where c is a constant	$\int 0\,dx=c$, where c is a constant
$\frac{d}{dx}(kx)=k,$ where k is a constant	$\int k\,dx=kx+c$ where c is a constant
$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right)=x^n$	$\int x^n dx=\frac{x^{n+1}}{n+1}+c, \quad n\neq -1$ (Power rule)
$\frac{d}{dx}\log x=\left(\frac{1}{x}\right)$	$\int \frac{1}{x}\,dx=\log\lvert x\rvert+c$
$\frac{d}{dx}(-\cos x)=\sin x$	$\int \sin x\,dx=-\cos x+c$
$\frac{d}{dx}(\sin x)=\cos x$	$\int \cos x\,dx=\sin x+c$
$\frac{d}{dx}(\tan x)=\sec^2 x$	$\int \sec^2 x\,dx=\tan x+c$
$\frac{d}{dx}(-\cot x)=\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x\,dx=-\cot x+c$
$\frac{d}{dx}(\sec x)=\sec x\tan x$	$\int \sec x\tan x\,dx=\sec x+c$
$\frac{d}{dx}(-\operatorname{cosec} x)=\operatorname{cosec} x\cot x$	$\int \operatorname{cosec} x\cot x\,dx=-\operatorname{cosec} x+c$
$\frac{d}{dx}(e^x)=e^x$	$\int e^x dx=e^x+c$
$\frac{d}{dx}\left(\frac{a^x}{\log a}\right)=a^x$	$\int a^x dx=\frac{a^x}{\log a}+c$
$\frac{d}{dx}(\sin^{-1}x)=\frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}}dx=\sin^{-1}x+c$
$\frac{d}{dx}(\tan^{-1}x)=\frac{1}{1+x^2}$	$\int \frac{1}{1+x^2}dx=\tan^{-1}x+c$

11_11th-Maths-Vol2-Unit11.indd 228 23-03-2019 10:16:07